W3(R) Length of surgeline and standpipe (m, ft). If input as 0, then the surgeline and standpipe are not modeled.

W4(R) Elevation drop of surgeline and standpipe (m, ft). This is the elevation drop from the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank) entrance to the injection point. A positive number denotes a decrease in elevation.

W5(R) Tank wall thickness (m, ft). This is not allowed to be 0.

W6(I) Heat transfer flag. If 0, heat transfer will be calculated. If 1, no heat transfer will be calculated.

W7(R) Tank density (kg/m^3, lb/ft^3). If 0, the density will default to that for carbon steel.

W8(R) Tank volumetric heat capacity (J/kg•K, Btu/lb•^{o}F). If 0, the heat capacity will default to that for carbon steel.

W9(I) Trip number. If 0 or if no number is input, then no trip test is performed. If nonzero then this must be a valid trip number, the operations performed are similar to those performed for a trip valve. If the trip is false, then the accumulator is isolated and no flow through the junction can occur. If the trip is true, then the accumulator is not isolated and flow through the junction will occur in the normal manner for an accumulator.

8.19.8 Fluidic Device Geometry, CCC2201

This cars is required if the geometry flag(W10) of accumulator geometry card CCC0101-0109 is set to 2 (cylindrical tank with fluidic device).

W1(R) Length of the stand pipe of fluidic device (m, ft). This value shlould be less than the length of tank.

W2(R) Reynolds number independent forward energy loss coefficient due to vortex motion. This quantity will be added to junction loss coefficient.

W3(R) Reynolds number independent reverse energy loss coefficient due to vortex motion. This quantity will be added to junction loss coefficient.

8.20 Sub-Domain Boundary Volume (SDBVOL) Component

A sub-domain boundary volume represents the three-dimensional cell that is connected to a one-dimensional cell.

This component is indicated by SDBVOL on card ccc0000. For major edits, minor edits, and plot variables, the volume in the sub-domain boundary volume components is numbered as ccc010000.

8.20.1 CARD ccc0101 through ccc0109, Sub-domain Boundary Volume Geometry Cards

This card (or cards) is required for a sub-domain boundary volume component. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Volume flow area (m2, ft2).

W2(R) Length of volume (m, ft).

W3(R) Volume of volume (m3, ft3).

W4(R) Azimuthal angle (degrees).

W5(R) Inclination angle (degrees). If a 1D cell is vertically connected to the top of the 3D cell represented by this sdbvol, this number is - 90.0.

If a 1D cell is vertically connected to the bottom of the 3D cell represented by this sdbvol, this number is 90.0.

If a 1D cell is horizontally connected to the 3D cell represented by this sdbvol, this number is 0.0 or 90.0. If the angle is 90.0, horizontal 1-D pipe is connected to this SDBVOL component using a cross flow junction.

W6(R) Elevation change (m, ft).

W7(R) Wall roughness (m, ft).

W8(R) Hydraulic diameter (m, ft).

W9(I) Volume control flags, tlpvbfe (See RELAP5/MOD3.2 Code Input Model).

8.20.2 CARD ccc0300, Sub-domain Boundary Volume Connection Cards

This card is required for a sub-domain boundary volume component.

W1(I) Channel number. This number is the channel number of the three-dimensional cell represented by this sdbvol.

W2(I) Mesh number. This number is the mesh number of the three-dimensional cell represented bythis sdbvol.

W3(I) Connection mode.

If a 1D cell is vertically connected to the top of the 3D cell represented by this sdbvol, this number is 1.

If a 1D cell is vertically connected to the bottom of the 3D cell represented by this sdbvol, this number is -1.

If a 1D cell is horizontally connected to the 3D cell represented by this sdbvol, this number is 0.

Additional three words (W4, W5 and W6) should be entered in case of horizontal connection (i.e., W3 is 0).

W4(I) Gap number 1.This number is the gap number facing the 1D/3D interface surface represented by this sdbvol.

This number is zero if there is no gap facing the 1D/3D interface surface.

W5(I) Gap number 2. This number is the left gap number normal to the 1D/3D interface represented by this sdbvol.

This number is zero if there is no gap normal to the 1D/3D interface surface.

W6(I) Gap number 3. This number is the right gap number normal to the 1D/3D interface.

This number is zero if there is no gap normal to the 1D/3D interface surface. The sequence of W5 and W6 is meaningless.

8.21 Multi-Dimensional Input

A multi-dimensional component is indicated by MULTID on Card CCC0000. This component defines a one-, two-, or three-dimensional array of volumes and the internal junctions connecting the volumes. The multi-dimensional component is described as a three-dimensional component but can be reduced to two or one dimensions by defining only one interval in the appropriate coordinate directions. The geometry can be either Cartesian (x,y,z) or cylindrical (r,θ,z). In cylindrical geometry, the r-direction can start at zero or nonzero, and θ can cover 360 degrees (i.e., a full circle) or can cover less than 360 degrees (wedge shape, semicircle, etc.).

An orthogonal, three-dimensional grid is defined by mesh interval input data in each of the three coordinate directions. The edges of the hydrodynamic volumes are defined by the grid lines. Given nx intervals in the x- or r-coordinate direction, ny intervals in the y- or θ-coordinate direction, and nz intervals in the z-coordinate direction, volumes are defined. The number of volumes in a threedimensional